UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN
Department of Electrical and Computer Engineering

ECE 310 Digital Signal Processing – Fall 2026

Homework 7

Profs. Kamalabadi, Snyder **Due: Friday, Oct. 16, 2025, 11:59pm on Gradescope**

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1. For each of the following LTI systems, compute and sketch the magnitude response $|H_d(\omega)|$ and phase response $\angle H_d(\omega)$ for $-\pi \leq \omega \leq \pi$.

   (a) $y[n] = x[n+2] + x[n-2]$

   (b) $y[n] = x[n] - x[n-4]$

   (c) $y[n] = x[n+2] + 2x[n] + x[n-2]$

2. A real-valued LTI system takes the following input $x[n]$

$$x[n] = 2 + \cos\left(\frac{\pi}{3}n - \frac{\pi}{3}\right) + \sin\left(\frac{3\pi}{4}n\right)$$

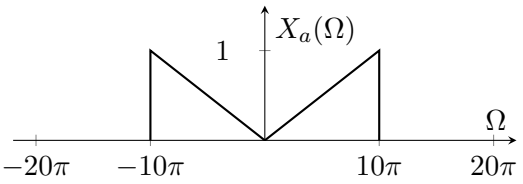
   and produces output $y[n]$:

$$y[n] = \frac{1}{2}\cos\left(\frac{\pi}{3}n\right) + \cos\left(\frac{3\pi}{4}n\right)$$

   Determine the system output to a new input signal $v[n]$ below.

$$v[n] = 1 + \sin\left(-\frac{\pi}{3}n\right) + \sin\left(\frac{3\pi}{4}n + \frac{\pi}{4}\right)$$

3. The continuous-time signal $x_a(t) \overset{\mathcal{F}}{\longleftrightarrow} X_a(\Omega)$ has the continuous-time Fourier transform shown below.



   The signal $x_a(t)$ is sampled with sampling period T to get the discrete-time signal $x[n] = x_a(nT)$ with DTFT $X_d(\omega)$.

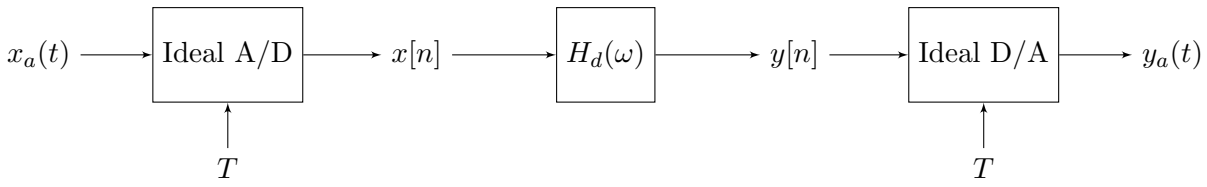
   (a) Determine the Nyquist rate of $x_a(t)$, i.e. the Nyquist sampling frequency f_{nyq}.

   (b) Sketch the DTFT $X_d(\omega)$ when $T = \frac{1}{40}$s for $-3\pi \leq \omega \leq 3\pi$.

   (c) Sketch the DTFT $X_d(\omega)$ when $T = \frac{1}{10}$s for $-3\pi \leq \omega \leq 3\pi$.

   (d) Sketch the DTFT $X_d(\omega)$ when $T = \frac{3}{20}$s for $-3\pi \leq \omega \leq 3\pi$.

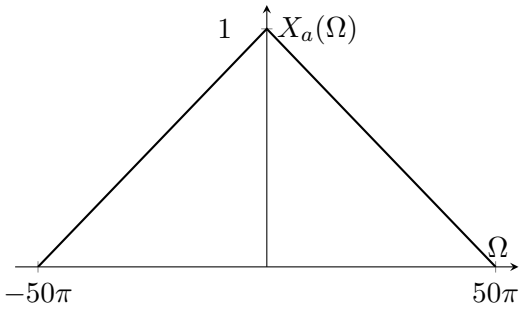
4. The below system consists of digital filter $H_d(\omega)$ fit between ideal A/D and D/A converters with shared sampling period T.



Suppose we know that $H_d(\omega)$ is given by the following low-pass filter.

$$H_d(\omega) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < |\omega| \leq \pi \end{cases}.$$

Let $X_a(\Omega)$, $X_d(\omega)$, $Y_d(\omega)$, and $Y_a(\Omega)$ be the Fourier transforms of $x_a(t), x[n], y[n]$ and $y_a(t)$, respectively. The CTFT $X_a(\Omega)$ is plotted below.



(a) Determine the Nyquist rate of $x_a(t)$, i.e. the Nyquist sampling frequency f_{nyq}.

(b) Sketch $X_d(\omega), Y_d(\omega)$ and $Y_a(\Omega)$ when $T = \frac{1}{50}$s. For $X_d(\omega)$ and $Y_d(\omega)$, sketch for $-3\pi \leq \omega \leq 3\pi$.

(c) Sketch $X_d(\omega), Y_d(\omega)$ and $Y_a(\Omega)$ when $T = \frac{1}{40}$s. For $X_d(\omega)$ and $Y_d(\omega)$, sketch for $-3\pi \leq \omega \leq 3\pi$.